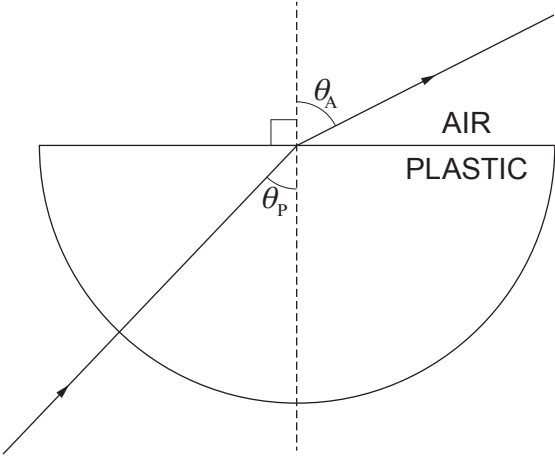


7. Rhodri carries out an experiment to determine the refractive index, n , of a plastic in the form of a block of semi-circular cross-section.

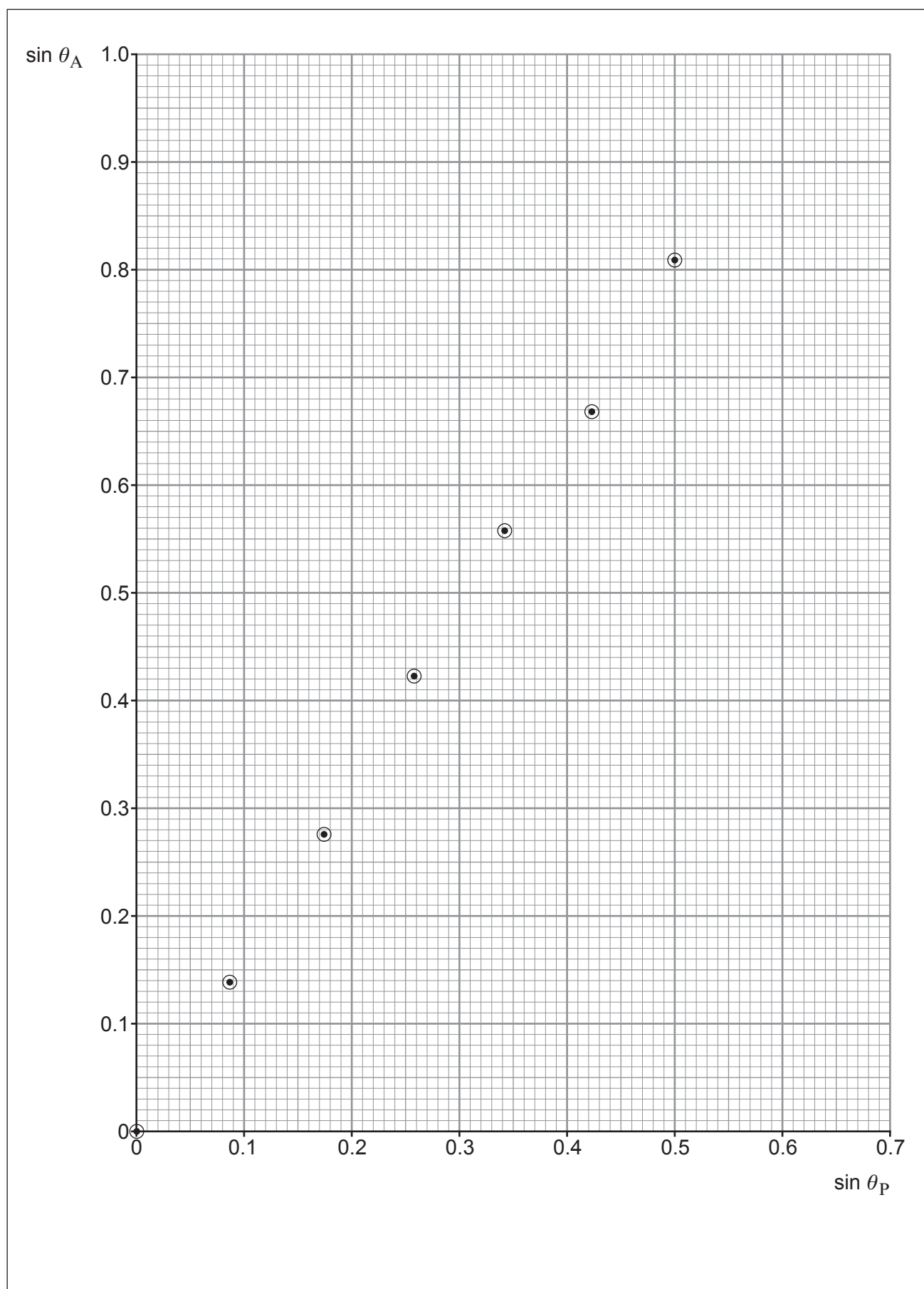


Rhodri places the block on paper and shines a narrow beam of light through the curved face, towards the circle centre. He measures angles θ_P and θ_A . He repeats the procedure for a range of angles of θ_P . His table of results is given, and his plot of $\sin \theta_A$ against $\sin \theta_P$.

$\theta_P / ^\circ$	$\sin \theta_P$	$\theta_A / ^\circ$	$\sin \theta_A$
0	0	0	0
5	0.087	8	0.139
10	0.174	16	0.276
15	0.259	25	0.423
20	0.342	34	0.559
25	0.423	42	0.669
30	0.500	54	0.809
35		66	
40		—	

- (a) State why no bending of the beam occurs as it enters the curved face of the block. [1]
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- (b) **Complete** the row of the table for $\theta_P = 35^\circ$ **and plot** the corresponding point on the grid opposite. [2]





- (c) (i) Discuss whether the graph supports the law that $\sin \theta_A$ is proportional to $\sin \theta_P$. [3]

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- (ii) Determine a value for the refractive index, n . [2]

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- (d) Rhodri could not obtain a value for θ_A when $\theta_P = 40^\circ$. Show why this had to be the case. [2]

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